

BIBLIOGRAPHY

- Ahlfors, L. V. 1979. *Complex analysis: an introduction to the theory of analytic functions of one complex variable*, International series in pure and applied mathematics, 3d ed edition. New York: McGraw-Hill.
- Altmann, S. L. 1989. Hamilton, Rodrigues, and the quaternion scandal. *Mathematics Magazine*, 62(5):291–308.
- Arianrhod, R. 2019. *Thomas Harriot: A Life in Science*. Oxford University Press.
- Arnol'd, V. I. 1989. *Mathematical Methods of Classical Mechanics*, volume 60 of *Graduate Texts in Mathematics*, 2nd edition. New York: Springer-Verlag.
- Arnol'd, V. I. 1990. *Huygens and Barrow, Newton and Hooke*. Basel: Birkhäuser Verlag. Pioneers in mathematical analysis and catastrophe theory from evolvents to quasicrystals, Translated from the Russian by Eric J. F. Primrose.
- Bak, J. and D. J. Newman 2010. *Complex Analysis*, Undergraduate texts in mathematics, 3rd ed edition. New York: Springer.
- Beardon, A. F. 1979. *Complex Analysis: the Argument Principle in Analysis and Topology*. Chichester: Wiley.
- Beardon, A. F. 1984. *A Primer on Riemann Surfaces*, volume 78. Cambridge: Cambridge University Press.
- Beardon, A. F. 1987. Curvature, circles, and conformal maps. *The American Mathematical Monthly*, 94(1):48–53.
- Beltrami, E. 1868a. Essay on the interpretation of non-Euclidean geometry. In *Stillwell* (1996). American Mathematical Society.
- Beltrami, E. 1868b. Fundamental theory of spaces of constant curvature. In *Stillwell* (1996). American Mathematical Society.
- Bivens, I. C. 1992. When do orthogonal families of curves possess a complex potential? *Mathematics Magazine*, 65(4):226–235.
- Boye, N. and S. Huggett 2011. Newton, the Geometer. *Newsletter of the European Mathematical Society*, (82):19–27.
- Boas, R. P. and H. P. Boas 2010. *Invitation to complex analysis*, MAA textbooks, 2nd ed edition. Mathematical Association of America.

- Bôcher, M. 1898. Note on poisson's integral. *Bull. Amer. Math. Soc.*, 4(9):424–426.
- Bôcher, M. 1906. On harmonic functions in two dimensions. *Proceedings of the American Academy of Arts and Sciences*, 41(26):577–583.
- Bohlin, K. 1911. Note sur le problème des deux corps et sur une integration nouvelle dans le problème des trois corps. *Bull. Astr.*, 28:113–119.
- Braden, B. 1985. Picturing functions of a complex variable. *The College Mathematics Journal*, 16(1):63–72.
- Braden, B. 1987. Polyá's geometric picture of complex contour integrals. *Mathematics Magazine*, 60(5):321–327.
- Braden, B. 1991. *Visualization in teaching and learning mathematics: a project*, volume no. 19, chapter : The Vector Field Approach in Complex Analysis. Mathematical Association of America.
- Brieskorn, E. and H. Knörrer 2012. *Plane Algebraic Curves*, Modern Birkhäuser classics. Basel: Birkhäuser.
- Carathéodory, C. 1937. The most general transformations of plane regions which transform circles into circles. *Bulletin of the American Mathematical Society*, 43(8):573–579.
- Carathéodory, C. 1964. *Theory of Functions of a Complex Variable*, 2nd english ed edition. New York: Chelsea Pub. Co.
- Chandrasekhar, S. 1995. *Newton's Principia for the Common Reader*. Oxford: The Clarendon Press, Oxford University Press.
- Courant, R. 1950. *Dirichlet's Principle, Conformal Mapping, and Minimal Surfaces*, volume 3. New York: Interscience Publishers.
- Coxeter, H. S. M. 1946. Quaternions and reflections. *The American Mathematical Monthly*, 53(3):136–146.
- Coxeter, H. S. M. 1967. *Proceedings of the International Conference on the Theory of Groups*, chapter : The Lorentz Group and the Group of Homographies, Pp. 73–77. New York: Gordon and Breach Science Publishers.
- Coxeter, H. S. M. 1969. *Introduction to Geometry*, 2nd edition. New York: Wiley.
- Coxeter, H. S. M. and S. L. Greitzer 1967. *Geometry Revisited*, volume 19. Random House.
- Davis, P. and H. Pollak 1958. On the analytic continuation of mapping functions. *Transactions of the American Mathematical Society*, 87(1):198–225.
- Davis, P. J. 1974. *The Schwarz Function and its Applications*, volume no. 17. Mathematical Association of America.

- de Gandt, F. 1995. *Force and Geometry in Newton's Principia*. Princeton, NJ: Princeton University Press. Translated from the French original and with an introduction by Curtis Wilson.
- do Carmo, M. P. 1994. *Differential Forms and Applications*, Universitext. Berlin: Springer-Verlag.
- Dray, T. 2015. *Differential Forms and the Geometry of General Relativity*. CRC Press, Taylor & Francis Group.
- Duffin, R. J. 1957. A note on Poisson's integral. *Quarterly of Applied Math*, 15:109–111.
- Earl, R. 2019. *Topology: A Very Short Introduction*, Very short introductions. Oxford University Press.
- Echols, W. H. 1923. Some properties of a skewsquare. *The American Mathematical Monthly*, 30(3):120–127.
- Eves, H. 1992. *Fundamentals of Modern Elementary Geometry*. Boston: Jones and Bartlett.
- Feynman, R. P. 1963. *The Feynman Lectures on Physics*. Reading, Mass.: Addison-Wesley Pub. Co.
- Feynman, R. P. 1966. Nobel lecture: "The Development of the Space-Time View of Quantum Field Theory". *Physics Today*, August:31–34.
- Feynman, R. P. 1985. *QED: the Strange Theory of Light and Matter*. Princeton, N.J.: Princeton University Press.
- Feynman, R. P. 1997. *Surely you're joking, Mr. Feynman!: Adventures of a Curious Character*, Norton paperback edition. New York: W.W. Norton.
- Finney, R. L. 1970. Dynamic proofs of Euclidean theorems. *Mathematics Magazine*, 43(4):177–185.
- Ford, L. R. 1929. *Automorphic Functions*, 1st edition. New York: McGraw-Hill book company, inc.
- Frankel, T. 2012. *The Geometry of Physics: an Introduction*, 3rd ed edition. Cambridge: Cambridge University Press.
- Fulton, W. 1995. *Algebraic Topology: A First Course*, volume 153. New York: Springer-Verlag.
- Gauss, C. F. 1827. *General Investigations of Curved Surfaces*, Translated from the Latin and German by Adam Hiltebeitel and James Morehead, 1965 edition. Raven Press, Hewlett, N.Y.
- Goodstein, J. R. 2018. *Einstein's Italian Mathematicians: Ricci, Levi-Civita, and the Birth of General Relativity*. AMS.

- Greenberg, M. J. 2008. *Euclidean and non-Euclidean Geometries*, 4th edition. New York: W.H. Freeman.
- Guicciardini, N. 1999. *Reading the Principia*. Cambridge: Cambridge University Press. The debate on Newton's mathematical methods for natural philosophy from 1687 to 1736.
- Guicciardini, N. 2009. *Isaac Newton on Mathematical Certainty and Method*, Transformations. Cambridge, Mass.: MIT Press.
- Henrici, P. 1991. *Applied and Computational Complex Analysis*, volume 3 of *Wiley classics library*, wiley classics library ed edition. New York: Wiley.
- Hilbert, D. 1902. *The Foundations of Geometry [Second English Edition of Tenth German Edition, translated by Leo Unger, Revised and Enlarged by Dr. Paul Bernays, published 1971]*. Chicago: The Open Court.
- Hilbert, D. 1952. *Geometry and the Imagination*. New York: Chelsea Pub. Co.
- Hilbert, D. 1965. *Gesammelte Abhandlungen*. Bronx, N.Y.: Chelsea Pub. Co.
- Hildebrandt, S. and A. Tromba 1985. *Mathematics and Optimal Form*. New York: Scientific American Library.
- Hoggar, S. G. 1992. *Mathematics for computer graphics*, volume 14. Cambridge: Cambridge University Press.
- Hopf, H. 1946–1956. *Differential Geometry in the Large: seminar lectures, New York University, 1946 and Stanford University, 1956 [Second Edition, 1983]*, volume 1000 of *Lecture notes in mathematics*. Berlin: Springer-Verlag.
- Horn, B. K. P. 1991. Relative orientation revisited. *J. Opt. Soc. Am. A*, 8(10): 1630–1638.
- Jones, G. A. and D. Singerman 1987. *Complex Functions: An Algebraic and Geometric Viewpoint*. Cambridge: Cambridge University Press.
- Kasner, E. 1912. Conformal geometry. In *Proceedings of the International Congress of Mathematicians, Vol. 2*.
- Kasner, E. 1913. *Differential-geometric Aspects of Dynamics*, American Mathematical Society. American Mathematical Society.
- Katok, S. 1992. *Fuchsian Groups*, Chicago lectures in mathematics series. Chicago: University of Chicago Press.
- Klein, F. 1881. *On Riemann's Theory of Algebraic Functions and their Integrals: A Supplement to the Usual Treatises ; translated from the German by Frances Hardcastle*. Mineola, NY: 2003, Dover Publications.
- Körner, T. W. 1988. *Fourier Analysis*. Cambridge: Cambridge University Press.

- Lanczos, C. 1966. *Discourse on Fourier series*. Edinburgh: Oliver and Boyd.
- Lange, R. and R. A. Walsh 1985. A heuristic for the poisson integral for the half plane and some caveats. *The American Mathematical Monthly*, 92(5): 356–358.
- Ludvigsen, M. 1999. *General Relativity: A Geometric Approach*. Cambridge: Cambridge University Press.
- Markushevich, A. 2005. *Theory of Functions of a Complex Variable*, number pt. 11 in AMS Chelsea Publishing Series, 2nd edition. AMS Chelsea Pub.
- Marsden, J., U. Marsden, M. Hoffman, U. Hoffman, and T. Marsden 1999. *Basic Complex Analysis*. W. H. Freeman.
- Maxwell, J. 1873. *A Treatise on Electricity and Magnetism*, A Treatise on Electricity and Magnetism 2 Volume Paperback Set. 2010, Cambridge University Press.
- Maxwell, J. 1881. *An Elementary Treatise on Electricity*, Cambridge Library Collection - Physical Sciences. 2011, Cambridge University Press.
- Milnor, J. 1982. Hyperbolic geometry: the first 150 years. *Bull. Amer. Math. Soc. (N.S.)*, 6(1):9–24.
- Misner, Thorne, and Wheeler 1973. *Gravitation*. San Francisco: W. H. Freeman [A marvellous new hardback edition was published by Princeton University Press in 2017. It contains a new introduction by Charles Misner and Kip Thorne, discussing exciting developments in the field since the book's original publication.].
- Mughal, A. and D. Weaire 2009. Curvature in conformal mappings of two-dimensional lattices and foam structure. *Proc. R. Soc., A*(465):219–238.
- Needham, T. 1993. Newton and the transmutation of force. *Amer. Math. Monthly*, 100(2):119–137.
- Needham, T. 1994. The geometry of harmonic functions. *Mathematics Magazine*, 67(2):92–108.
- Needham, T. 2014. Visual Differential Geometry and Beltrami's Hyperbolic Plane. In *The Art of Science: From Perspective Drawing to Quantum Randomness*, R. Lupacchini and A. Angelini, eds., Pp. 71–99. Springer International Publishing.
- Needham, T. 2021. *Visual Differential Geometry and Forms: A Mathematical Drama in Five Acts*. Princeton, N.J.: Princeton University Press.
- Nehari, Z. 1952. *Conformal mapping*. New York: Dover Publications.
- Neumann, C. 1884. *Vorlesungen über Riemann's Theorie der Abel'schen Integrale*. Leipzig : B. G. Teubner.

- Nevanlinna, R. H. and V. Paatero 2007. *Introduction to Complex Analysis*, 2nd edition. Providence, R.I.: AMS Chelsea Pub./ American Mathematical Society.
- Newton, I. 1670. *The Mathematical Papers of Isaac Newton*, volume III. Cambridge: 1981, Cambridge U.P.
- Newton, I. 1687. *The Principia: Mathematical Principles of Natural Philosophy*, 1999 edition. Berkeley, CA: University of California Press. A new translation by I. Bernard Cohen and Anne Whitman, assisted by Julia Budenz, Preceded by "A guide to Newton's *Principia*" by Cohen.
- Nikulin, V. V. and I. R. Shafarevich 1987. *Geometries and Groups*, Universitext. Berlin: Springer-Verlag.
- Nitsche, J. C. C. 1989. *Lectures on Minimal Surfaces*. Cambridge: Cambridge University Press.
- Ogilvy, C. S. 1990. *Excursions in Geometry*. New York: Dover Publications.
- O'Neill, B. 2006. *Elementary Differential Geometry*, Revised 2nd edition. Amsterdam: Elsevier Academic Press.
- Osgood, W. 1928. *Lehrbuch der Funktionentheorie*. Teubner.
- Penrose, R. 1978. The geometry of the universe. In *Mathematics today: twelve informal essays*, L. A. Steen, ed., Pp. 83–127. New York: Vintage Books.
- Penrose, R. 2005. *The Road to Reality*. New York: Alfred A. Knopf Inc. A complete guide to the laws of the universe.
- Penrose, R. and W. Rindler 1984. *Spinors and Space-Time*. Cambridge: Cambridge University Press.
- Perkins, F. W. 1928. An intrinsic treatment of Poisson's Integral. *American Journal of Mathematics*, 50(3):389–414.
- Poincaré, H. 1985. *Papers on Fuchsian Functions*. New York: Springer-Verlag.
- Pólya, G. 1954. *Mathematics and Plausible Reasoning*. Princeton, N.J.: Princeton University Press.
- Pólya, G. and G. Latta 1974. *Complex Variables*. New York: Wiley.
- Prasolov, V. V. 1995. *Intuitive Topology*, volume v. 4. American Mathematical Society.
- Reech 1858. *Journal de l'École Polytechnique*, 27:169–178.
- Richeson, D. S. 2008. *Euler's Gem: The Polyhedron Formula and the Birth of Topology*. Princeton, N.J.: Princeton University Press.
- Schutz, B. F. 1980. *Geometrical Methods of Mathematical Physics*. Cambridge: Cambridge University Press.

- Schutz, B. F. 2022. *A First Course in General Relativity*, 3rd edition. Cambridge: Cambridge University Press.
- Schwarz, H. 1972a. *Gesammelte Mathematische Abhandlungen*, volume II, Pp. 144–171. Chelsea Publishing.
- Schwarz, H. 1972b. *Gesammelte Mathematische Abhandlungen*, volume II, P. 360. Chelsea Publishing.
- Shapiro, H. S. 1992. *The Schwarz Function and its Generalization to Higher Dimensions*, volume v. 9. New York: Wiley.
- Shaw, W. T. 2006. *Complex Analysis with Mathematica*. Cambridge, UK: Cambridge University Press.
- Siegel, C. L. 1969. *Topics in Complex Function Theory*, volume 25. New York: Wiley-Interscience.
- Sommerville, D. 1958. *The Elements of Non-Euclidean Geometry*. Dover.
- Springer, G. 1981. *Introduction to Riemann Surfaces*, 2nd ed edition. New York: Chelsea Pub. Co.
- Stewart, G. W. 1993. On the early history of the singular value decomposition. *SIAM Review*, 35(4):551–566.
- Stewart, I. and D. O. Tall 2018. *Complex Analysis: The Hitch Hiker's Guide to the Plane*, second edition. Cambridge University Press.
- Stillwell, J. 1992. *Geometry of Surfaces*, Universitext. New York: Springer-Verlag.
- Stillwell, J. 1994. *Elements of Algebra: Geometry, Numbers, Equations*. New York: Springer-Verlag.
- Stillwell, J. 1996. *Sources of Hyperbolic Geometry*, volume 10 of *History of Mathematics*. Providence, RI: American Mathematical Society.
- Stillwell, J. 2010. *Mathematics and its History*, third edition. New York: Springer.
- Thorne, K. S. and R. D. Blandford 2017. *Modern Classical Physics: Optics, Fluids, Plasmas, Elasticity, Relativity, and Statistical Physics*. Princeton University Press.
- Thurston, W. P. 1997. *Three-Dimensional Geometry and Topology*, volume 35. Princeton, N.J.: Princeton University Press.
- Trefethen, L. N. 1986. *Numerical Conformal Mapping*. Amsterdam: North-Holland.
- Vitelli, V. and D. R. Nelson 2006. Nematic textures in spherical shells. *Phys. Rev. E*, 74:021711.
- Waerden, B. L. v. d. 1985. *A history of algebra: from al-Khwārizmī to Emmy Noether*. Berlin: Springer-Verlag.

Wald, R. M. 1984. *General Relativity*. Chicago: University of Chicago Press.

Westfall, R. S. 1980. *Never at Rest*. Cambridge: Cambridge University Press. A biography of Isaac Newton.

Whyburn, G. 1955. *Introductory Topological Analysis*, number v. 783, no. 22 in *Introductory Topological Analysis*. University of Michigan Press.

Whyburn, G. 2015. *Topological Analysis*, Princeton Legacy Library, second edition. Princeton University Press.

Woodward, L. M. and J. Bolton 2019. *A First Course in Differential Geometry: Surfaces in Euclidean Space*. Cambridge University Press.

Yaglom, I. M. 1988. *Felix Klein and Sophus Lie: evolution of the idea of symmetry in the Nineteenth Century*. Boston: Birkhäuser.

INDEX

- 2-spinors, 179
- $\asymp$ -notation, xvi
- Abel's convergence theorem, 78
- absolute convergence, 76, 77
- acceleration, 299, 478
- Ahlfors, L. V., 175, 230
- algebra
 - Cotes' work on, 25
 - cubics, 5, 51, 66
 - cyclotomic polynomials, 57
 - Descartes' Factor Theorem, 26
 - Fundamental Theorem of, 26, 27, 404
 - quaternions, 49, 329–333, 374, 375
- algebraic branch point, 102
- algebraic multiplicity, 394–396, 398
- amplification, 218, 224
- amplify, to, 218, 219, 224
- amplitwist
 - as verb, 219
 - defines analytic mappings, 223
 - definition of, 218, 224
 - German precedent for, xxvii
 - of $\exp(z)$, 262–264
 - of $\log(z)$, 252–253
 - of z^a , 260–262
 - of complex inversion, 229, 241
 - of inverse mapping, 255
 - of power series, 259
 - rules for calculating, 254–256
 - terminology illustrated, 219
- analytic continuation, 281
 - preservation of identities, 286
 - uniqueness, 284–286
 - via reflections, 286–292
- analytic function, 223–225
 - and Riemann sphere, 230
 - equivalent definitions of, 227–230
- angle of parallelism, 347, 381
- angular excess, 305–307
 - additivity of, 374
 - in hyperbolic plane, 357–359
 - on general surface, 312, 314, 383
 - on sphere, 316–317
- angular momentum, 274, 279
- animation, computer, 332, 576
- annular Laurent series, 501–505
- anticonformal mapping, 147
- antiderivative, 462–464
- antipodal mapping, 321, 533
- antipodal points
 - and inversion, 167, 205, 206
 - as intersection of great circles, 318
 - Borsuk-Ulam Theorem, 425
 - definition of, 167
 - non-uniqueness of great circle between, 317
 - orthogonal in $\mathbb{C}^2$, 183
 - stereographic images of, 167
 - used to define axis of rotation, 318
 - used to find sphere rotation formula, 326
- antipodal triangle, 316
- Apollonius, xv
- Apollonius, circles of, 197
- approximation
 - of $(1 + \epsilon)$, 93
 - of $\exp(z)$, 91–93, 112
 - of area under curve, 430–436
 - of complex integral, 437–440
 - of power series, 78, 80, 122, 127, 128, 257
 - of vector field locally, 553, 554
- Archimedes, 121
- area
 - as integral of $\bar{z}$, 448–450, 550–551
 - hyperbolic, 357–359
 - non-Euclidean, 306
 - of polygon, 33
 - signed, 31
 - spherical, 316–317

- Area Theorem, Bieberbach's, 483
- areal speed, 274, 299
- Argand, J. R., 2, 3
- Argument Principle
 - for polynomials, 393–394
 - for rational functions, 414–416
 - generalized, 415
 - topological, 399–401
- Arianrhod, R., 161
- arithmetic
 - equivalence of symbolic and geometric, 8–11
- Arnol'd, V. I.
 - as champion of Newton's geometrical method, xviii
 - dual force fields, 276, 277, 280
 - Kasner–Arnol'd Theorem, 280
- astronomy, 156
- asymptotic
 - h-lines, 346
 - triangle, 357–359
- atlas, geographic
 - and the metric of a surface, 322
 - definition of, 158
 - for pseudospherical surfaces, 341
 - stereographic and Mercator maps, 294
- attractive fixed point, 187
- automorphism
 - definition of, 199
 - of disc, 199–204, 211, 212
 - connections with non-Euclidean geometry, 315, 367
 - uniqueness, 201
 - via physics, 624–626
 - of upper half-plane, 212
 - connections with non-Euclidean geometry, 349
- average
 - complex, 483
 - explanation of Cauchy's Formula, 486–487
 - interpretation of Green's formula, 645
 - interpretation of Laplacian, 586
 - interpretation of Poisson's formula, 631
 - of harmonic function, 487, 581
 - over circle, 121–124
 - over polygon, 118–121
- axiom
 - hyperbolic, 304
 - parallel, 304
 - spherical, 304
- Bôcher's formula, 641
- Bak, J. and Newman, D. J., 506, 618
- basis vectors, 330
- Beardon, A. F.
 - and Pick's curvature formula, 298
 - and Schwarzian derivative, 297, 481
 - and topological approach to infinite differentiability, 485
- Beltrami, E.
 - conformal "Poincaré" model, 359–363
 - discovery of hyperbolic models, xii
 - hemisphere model, 367–369
 - hyperbolic plane, 339–342
 - interpretation of non-Euclidean geometry, 313
 - projective "Klein" model, 367
 - pseudosphere model, 307
- Beltrami–Poincaré disc model, 359–363
 - motions of, 363–367
- Beltrami–Poincaré models of the hyperbolic plane, 342
- Beltrami–Poincaré upper half-plane, 349
- Beltrami–Poincarites, xii, 350
- Bendixson's Formula, 533
- Bernoulli, 498
 - James, 68, 126
 - John, 132
- Bieberbach's Area Theorem, 483
- bilinear mappings, 137
- binary rotation, 331
- Binomial Theorem
 - and the series for $\log(z)$, 113
 - and the series for e^z , 92
 - convergence and branch points, 106–110
 - Euler's derivation of, 129
 - for negative integer powers, 128, 129
 - Newton's and Euler's use of, 82
 - using complex numbers, 56
- Bivens, I. C., 592, 595
- Bôcher, M., 640
- Bohlin, K., 277
- Bolyai, J., 307, 347
- Bolyai–Lobachevsky Formula, 347

- geometric proof, 348
- Bombelli, R., 1, 4
- Borsuk-Ulam Theorem, 425
- Braden's Theorem, 575
- Braden, B., xxi, 549, 556, 575
- branch cut, 104, 105, 109, 110, 131
- branch of multifunction, 103–106
- branch point, 101
 - algebraic, 102
 - as obstacle to power series, 107
 - at infinity, 164
 - logarithmic, 102
 - order of, 102
 - related to critical point, 233
- Brouwer's Fixed Point Theorem, 404–405, 424
- Burnett-Stuart, G.
 - author's debt to, xxx
- calculus (real)
 - and complex arithmetic, 22–25
 - Newton's three versions of, xxiv
- Cardano, G., 1, 4, 5
- Cardano-Tartaglia formula, 51
- cardioid, 125
- Cassini, G., 67
- Cassinian curve, 67–70, 125, 126, 576
- Cauchy's Formula, 485
 - derived via averaging, 486–487
 - derived via Cauchy's Theorem, 488–489
 - physical explanation of, 556
 - used to derive existence of Laurent series, 502–505
 - used to derive existence of Taylor series, 491–493
 - used to derive infinite differentiability, 489–490
- Cauchy's Theorem
 - and antiderivatives, 464
 - as nonlocal manifestation of local amplitwist, 442
 - for a general loop, 472–475
 - geometric explanation of, 467–471
 - informal introduction to, 430
 - used to derive Deformation Theorem, 453, 454
- Cauchy-Riemann equations
 - explained visually, 245–247
- polar form, 247–248
 - used to derive $\log(z)$, 249–252
 - via the Jacobian, 237–238
- Cauchy-Hadamard Theorem, 84
- Cayley, A., 332, 426
- celestial mechanics, 274–280
- central dilation, 45
- centroid, 134, 135, 487
 - of regular n -gon, 116, 135
 - of triangle, 53
 - of triangle constructed via equilateral triangles, 54
 - used to explain Gauss's Mean Value Theorem, 115–121
- chain rule of calculus, 254, 297
- Chandrasekhar, S., 276
 - and $\asymp$ -notation, xvi
 - on Newton's *Principia*, 23
- characteristic equation, 181, 422
- circle at infinity, 341, 343
- circle of convergence, 78, 299
- circle of curvature, 270, 300, 336, 435
- circles, touching, 153–154, 208
- circulation of vector field, 541
 - Stokes's theorem for, 547
 - vanishing of, xxi, 541, 545, 549, 557
 - work interpretation of, 539–542
- classification
 - of Euclidean motions, 38–41
 - of hyperbolic motions, 348–357
 - of Möbius transformations, 184–192, 194
 - of singular points, 518
 - of singularities, 416–418, 503
 - of spherical motions, 317–321
 - of surfaces, 526
- complex curvature
 - additional curvature formula (in Ex. 11), 650
 - advanced properties, 590–597
 - and Kasner's Invariant (in Ex. 10), 650
 - applied to dual central force fields, 279–280
 - applied to the geometry of harmonic functions, 590–594
 - curl of, 595
 - curvature of streamlines and equipotentials in terms of complex potential, 595
 - definition of, 271
 - elementary properties, 270–274

- complex curvature (*Continued*)
 - further geometry of, 595–597
 - geometric interpretation of, 271
 - geometric interpretation of real and imaginary parts, 272
 - interpretation in terms of rotation and expansion of infinitesimal shapes, 273
 - of various mappings (in Ex. 17), 295
 - VCA introduced both the concept and the name, xxi
- complex gradient operator, 588–589, 593, 649
- complex inversion, 139–150
 - amplitwist of, 229, 241
 - as rotation of Riemann sphere, 163
 - integral of, and winding number, 442–446, 551–553
 - of hyperbola into lemniscate, 211
- complex numbers, 1–8
- complex potential, 569–571
- confocal conics, 67, 100, 612
- conformal invariant, 296, 650
- conformal mapping, 147, 227–230
- congruence, 34
- conjugate
 - harmonic function, 577
 - transpose matrix, 183
- conjugate transpose, 183
- conjugation mapping
 - analytic fake of, 289
 - area as integral of, 447–450, 550–551
 - as reflection of Riemann sphere, 162
- connected set, 103
- conservation of energy, 274, 541
- conservative vector field, 541, 567–569
- contour, 429
- convergence
 - absolute, 76, 77
 - circle of, 78, 299
 - disc of, 75–78
 - radius of, 71, 73, 74, 78
 - tests for, 83–86
 - uniform, 79
- convex hull, 116, 134, 294
- convex region, 295
- cosine, 94–100
- Cotes' result, 25–30
- Cotes, R., 25–27, 29
- Courant, R., 618
- Coxeter, H. S. M., 48, 137, 375
- critical point
 - at infinity, 164
 - defined for non-analytic case, 399
 - definition of, 164, 231, 534
 - degrees of crushing, 231–232
 - geometric significance, 421, 425
 - Lucas' Theorem, 294
 - order of, 232
 - physical significance, 571, 615, 616, 650
 - relation to branch point, 233
 - use in counting preimages, 394–396
- cross product, 31
 - discovered via quaternions, 330
- cross-ratio, 174–177
 - hyperbolic distance in terms of, 382
 - projective invariance, 209
 - six possible values of, 210
- crossing rule for winding numbers, 388
- crosspoint, 518
 - index of, 519
- cube roots, 100–106
- cubic
 - Cardano-Tartaglia solution of, 51
 - geometry of Cardano-Tartaglia formula, 66
 - role in discovery of complex numbers, 5
 - Viète solution of, 66
- curl of vector field, 543
 - geometric formula for, 545, 574, 593
 - vanishing of, xxi, 545, 549, 557
- curvature, 210, 269
 - applied to celestial mechanics, 278–280
 - of harmonic equipotentials, 590–594
 - principal, 312
 - related to divergence and curl, 543–545
 - related to Schwarz function, 299
 - related to Schwarzian derivative, 298
 - transformation by analytic function, 267–270, 650
 - used to estimate error of Riemann sum, 434–436
- cyclotomic polynomial, 57
- Davis, P. J. and Pollak, H., 289
- Deformation Theorem
 - for analytic function, 453–454

- for flux, 547
- for work, 547
- Hopf's, 388–393
- degree of a mapping, 390, 391
- degrees of freedom
 - in Riemann's Mapping Theorem, 204, 613
 - of dipole mapping, 620
 - of disc automorphisms, 199–201
 - of Green's function, 627
 - of Singular Value Decomposition, 236
 - of upper half-plane automorphisms, 212
 - using flows to construct mappings, 617–618
- Desargues, G., 175
- Descartes' Factor Theorem, 26
- determinant
 - of Jacobian, 240, 398, 421, 649
 - of Möbius transformation, 178, 180
- differential geometry
 - angular excess interpretation of curvature, 312
 - Beltrami's use of in hyperbolic geometry, 307
 - conformal curvature formula, 380
 - constant negative curvature of pseudosphere, 335–336, 357
 - Further Reading, 373
 - Gauss's 1827 breakthrough, 311
 - metric of a surface, 323
 - parallel transport, 383
- dilative rotation, 45, 50
- dipole
 - as fusion of source and sink, 558
 - at infinity, 560
 - complex potential proof of circular streamlines, 572
 - dipole moment of, 559, 560
 - dual of, 578, 649
 - fused to create quadrupole, 559
 - image of, 604, 607
 - index of, 519
 - inserted into uniform flow, 572
 - on Riemann sphere, 525, 609
 - Pólya vector field, 557
 - superposition of multipoles, 561, 562
 - used to construct Riemann mapping, 618–620
 - vanishing integral of, 557
- Dirichlet's problem, 630–648
- disc of convergence, 75–78
- disconnected set, 103
- distributive property, 6, 10
- divergence of vector field, 543
 - Gauss's theorem on, 547
 - geometric formula for, 544, 574, 593
 - vanishing of, xxi, 545, 549, 557
- dot product, 30
 - discovered via quaternions, 330
 - in $\mathbb{C}^2$, 182
- double pole, 417
- douplet, 518
- doughnut
 - and lemniscate, 70
 - and Riemann surfaces, 230
 - is called a torus, 526
 - spiral sections of Perseus, 70
 - vector field on, 527
- dual
 - force laws, 280
 - harmonic, 577
- Duffin, R. J., 637
- Earl, R, 531
- earthquake, location of, 38
- Echols, W. H., 483
- eigenvalue, 180–182
 - interpretation of multiplier, 192–194
- eigenvector, 180–182, 192–194
- Einstein, A.
 - and non-Euclidean geometry, 307, 379
 - General Relativity, 303
 - Georg Pick's influence on, 411
 - Lorentz transformations as Möbius transformations, 139
 - Penrose's 2-spinors, 179
 - spacetime interval, 138
 - Special Relativity, 137, 138
- electricity, flow of, 513–515
- electrostatic field
 - as equilibrium state, 603
 - behaves like ideal fluid, 516
 - dual of, 603
 - energy contained in, 570, 643
 - Further Reading, 604
 - interpretation of Green's function, 623

- electrostatic field (*Continued*)
 - method of images, 604–611
 - of doublet, 516, 518
 - potential of, 567
- elliptic integral, 68, 131, 243
- elliptical orbits, 274–278
- energy
 - conservation of, 274, 541
 - electrostatic, 570, 643
 - total, 274, 280
 - used to define potential function, 567
 - work, 539–543
- epicycloid, 478
- equidistant curve
 - angular definition of hyperbolic distance, 361
 - definition, 353
 - in Beltrami–Poincaré disc, 361
- equipotentials, 567
 - l-work tube, 567, 568
 - harmonic, 590–594
- equivalence relation, 36
- Erlangen Program, 37
- Euclidean geometry, 303
 - motions of, 38–41
 - similarities of, 44–48
 - spatial complex numbers?, 48–50
 - Three Reflections Theorem, 42
 - through the eyes of Felix Klein, 33–38
- Euler characteristic, 527
- Euler’s formula, 11–15
- Euler’s Polyhedral Formula, 534
- Euler, L., 501
 - and Binomial Theorem, 82
 - and birth of complex numbers, 2
 - and principal curvatures, 312
 - discovery of Euler’s formula, 11
 - proof of Binomial Theorem, 129
 - sum of reciprocals of squares, 498, 509
- Euler-Lhuillier Formula, 534
- exponential function, 88–94
- extended complex plane, 157
- extrinsic geometry, 309
- Faraday, M., 515
 - k-flux tube terminology, 563
- Feynman, R. P.
 - bet by, 496
 - Nobel lecture, xxiii
 - on complex numbers in quantum mechanics, xxvii
 - path integrals, 438
 - visualizing circulation, 541
- fiducial vector field, 528
- Finney, R. L., 19
- fixed point, 171
 - at infinity, 172
 - attractive, 187
 - Brouwer’s Theorem, 404–405, 424
 - repulsive, 187
- flows, 513–518
 - around obstacles, 598–614
 - dual, 577–580
 - on curved surfaces, 525–531
- flux of vector field, 516, 537
 - k-flux tube, 563
 - Gauss’s theorem for, 547
 - local density of, 543
 - vanishing of, xxi, 539, 545, 549, 557
- foci of Cassinian curves, 68
- force fields, 274, 515
 - conservative, 541
 - dual, 280
 - linear and inverse-square, 274–278
- Ford, L. R., 186, 210
- Fourier series
 - complex unification with Taylor series, 86–88
 - examples derived from power series, 130, 134
 - representation of discontinuous functions, 634
- Fourier, J.
 - discovery of Fourier series, 86
- Fundamental Theorem of Algebra, 26, 27, 404
- Fundamental Theorem of Calculus, 458–462
- Gauss’s Divergence Theorem, 547
- Gauss’s Mean Value Theorem
 - and Dirichlet’s Problem, 630
 - for power series, 124, 646
 - proof of, 581
 - related to Laplacian, 586

- second proof of, 587
- statement of, 487
- Gauss, C. F., 184
 - complex plane, 2, 3
 - differential geometry, 308, 311, 312
 - non-Euclidean geometry, 306, 307
 - quaternions, 50
 - rotations of sphere, 315
- Gaussian curvature, 311–313
 - formula in terms of metric, 380
 - of hyperbolic plane, 357
 - of pseudosphere, 335–336
 - of sphere, 316
 - surfaces of constant, 313–315
- Gaussian integers, 54
- genus of surface, 526
- geodesic, 308
- geometry
 - Erlangen Program, 37
 - extrinsic, 309
 - intrinsic, 310
 - non-Euclidean, 33–38, 303–315
 - on curved surface, 307–309
 - projective, 37, 175, 179
 - via complex arithmetic, 18–22
- glide reflection, 58
- gradient operator, 565–567
 - complex, 588–589, 593, 649
- gravitational orbits, 274–280
- Green's formula, 643
- Green's Function, 626–630, 650, 651
- Green's General Formula, 643–648
- group, 37
- Hamilton, W. R., 49, 329, 330, 332, 333
- harmonic family of curves, 590–594
 - curvature test for, 591
 - definition of, 590
 - geometric test for, 591
- harmonic function
 - conformal invariance of, 583–586
 - conjugate, 577
 - Dirichlet's Problem for, 630–648
 - dual of, 577
 - maxima and minima of, 407
 - potential function, 569
 - stream function, 567
- Harnack's inequality, 651
- Harriot, T.
 - angular excess of spherical triangle, 316
 - conformality of stereographic projection, 161
- Hawking, S., 161
- heat flow
 - and Green's function, 623, 626–648
 - around an obstacle, 598
 - steady state, 515
 - temperature as potential function, 567
- helix, 298
- Henrici, P., 614, 618
- Hexlet, Soddy's, 209
- Hilbert, D., 342, 372, 618
- holes, 103, 523, 526
- holomorphic, 223
- homogeneous coordinates, 178–180, 183, 193, 376
- homographic transformations, 137
- Hopf's Degree Theorem, 388–393
- Hopf, H., 529
- horizon, 341
- horocycle, 351
- horosphere, 372
- Hurwitz polynomial, 422
- Huygens
 - influence on Newton, xv
- Huygens, C., 334, 357
- hydrodynamics, 567
- hydrogen atom, 506
- hyperbolic functions, 95–100
- hyperbolic geometry, 305, 307, 313, 315
 - absolute unit of length, 306
 - angle of parallelism, 347
 - angular excess, 357–359
 - angular excess proportional to area, 306
 - Beltrami–Klein model, 367, 377, 378
 - Beltrami–Poincaré disc model, 359–363
 - Bolyai–Lobachevsky Formula, 347
 - Bolyai–Lobachevsky Formula (geometric proof), 348
 - construction for distance, 361
 - curvature of, 307
 - equidistant curve, 353, 361
 - hemisphere model, 367–369
 - horizon, 341

hyperbolic geometry (*Continued*)

- horocycle, 351
 - horosphere, 372
 - hyperbolic plane, 339–342
 - lines and reflections, 342–347
 - maximum area of triangle, 306
 - motions in terms of reflections, 353–357
 - motions of, 348–353
 - Poincaré upper half-plane model, 342
 - Schwarz–Pick Lemma, 412
 - triangles have negative angular excess, 306
 - unifies solutions of Dirichlet’s Problem, 647
 - via tractrix and pseudosphere, 333–339
- hyperbolic Möbius transformation, 174, 187

ideal points, 341

identities

- hyperbolic, 95
- preservation of, 286
- trigonometric, 16–17

identity matrix, 178

Identity Theorem, 79

identity transformation, 19, 36

imaginary number, 2

incompressibility, 516

index of singular point

- Bendixson’s Formula, 533
- defined in the plane, 519
- defined on a curved surface, 528
- Index Theorem, 523–525
- of Pólya vector field, 548
- Poincaré’s method of finding, 522–523
- Poincaré–Hopf Theorem, 526, 529–530

induced mapping, 162

infinitesimal, 23

infinity, point at, 157

“inside” (of closed curve), 386–387

integral, complex

- as limit of Riemann sum, 437–440
- parametric evaluation of, 466–467
- path-independence of, 442, 462–464

integration by parts, 481

interior (of closed curve), 386–387

interval of convergence, 71

intrinsic

- geometry, 310

invariant curve, 42

inversion in a circle, 139–142

- geometric constructions of, 144–147, 205
- illustrative applications of, 153–157
- mechanical construction of, 205
- preservation of angles, 147–149
- preservation of circles, 142–144
- preservation of symmetry, 149–150

inversion in a sphere, 150–153, 205

- local geometry of, 243
- relation to stereographic projection, 161
- role in hyperbolic geometry, 367–373

involutory transformation

- and classification of $z \mapsto (1/z)$, 211
- definition of, 141
- geometric inversion as, 143
- hyperbolic half-turn of the disc, 202, 203, 365
- using matrix representation of M_a , 212

irrotational vector field

- definition of, 542
- generated by a given potential, 581
- has harmonic potential function
 - if sourceless, 567
- potential function, 567
- unique flow round obstacle, 600
- used to construct conformal mapping, 616
- vanishing curl and Stokes’s theorem, 545–547

isogonal mapping, 148

isolated singularity, 503

isometric circle, 210, 242

isometry, 35

isotherm, 623, 627–629, 637

Jacobian matrix

- and topological multiplicity, 398
- definition of, 217–218
- describes the local transformation, 235
- determinant via ∂_z and $\partial_{\bar{z}}$, 649
- eigenvalues and the SVD, 421
- in non-analytic example, 422
- in three dimensions, 421
- local area magnification is
 - (amplification)², 240
- of $z \mapsto (1/z)$, 241
- used to derive Cauchy–Riemann equations, 237

- Kasner's Invariant, 650
- Kasner, E., 280
- "angle" between touching curves, 296
- dual force fields, 277
- Kasner–Arnol'd Theorem, 277, 280
- Kepler's equation, 71
- Klein, F.
- Erlangen Program*, 33–38
- and amplitwist terminology, xxvii
- Beltrami–Klein model of the hyperbolic plane, 342
- classification of Möbius transformations, 184
- monograph on Riemann surfaces, xxvii, 62, 530
- projective model of hyperbolic plane, xii
- Koebe, P., 614
- Kubrick, S.
- 2001: *A Space Odyssey*, xv
- Lambert, J. H., 306
- Lanczos, C., 86
- Lange, R. and Walsh, R. A., 643
- Laplace's equation, 293
- Laplacian operator, 293
- complex gradient form of, 589
- conformal invariance of, 584–586
- geometric interpretation of, 586–587
- in terms of ∂_z , 649
- Laurent series, 456, 477
- and poles, 493–494
- annular, 501–505
- classification of singularities, 503
- Legendre polynomials, 506
- Legendre, A., 534
- Leibniz, xv
- Leibniz's rule, 127
- Leibniz, G., 1
- series for π , 113
- lemniscate
- as Cassinian curve, 69
- as image of hyperbola under inversion, 211
- as spiric section of Perseus, 70
- as streamline and equipotential of quadrupole, 576
- definition of, 68
- James Bernouilli's *standard lemniscate*, 126
- lemniscatic integral, 131, 242
- self-intersects at right angles, 125
- lemniscatic integral, 68, 131, 243
- Liebmann, H., 137, 321
- limit rotation, 350
- linear transformation
- geometry of, 235–237
- Jacobian, 217
- Möbius transformation as, 179
- matrix representation of, 177
- of spacetime, 138
- linear-fractional transformations, 137
- Liouville's Theorem, 409–411, 506, 651
- Liouville, J., 243
- Lobachevsky, N., 307, 347
- local degree, 397
- logarithm, 110–112
- and analytic "rigidity", 249–252
- and Mercator map projection, 294
- as integral, 465–466
- definition of general powers, 113–115
- power series for, 112–113
- principal value of, 111
- visual differentiation of, 252–253
- logarithmic branch point, 102, 164
- logarithmic power series, 113, 132, 133
- loop
- "inside" of, 386–387
- simple, 386
- winding number of, 385
- Lorentz transformations, 138
- classification of, 172
- loxodromic Möbius transformation, 174, 187
- Lucas' Theorem, 294
- Mādhava, of Sangamagrāma, 113
- Macdonald's Theorem, 424, 425
- matrix
- conjugate transpose, 183
- identity, 178
- Jacobian, 217
- normalized, 177
- of Möbius transformation, 177–178
- unitary, 183
- Maximum-Modulus Theorem, 405–406
- Maxwell, J. C.
- k-flux tube terminology, 563

- Maxwell, J. C. (*Continued*)
 and Macdonald's Theorem, 426
 energy interpretation of k-cell, 569
 geometric approach to electrostatics, 604
 inversion is the only conformal mapping of
 3D space, 243
 mathematically exploited Faraday's field
 lines, 515
 physical explanation of Green's formula,
 643
 physically distinct interpretations of
 potential, 624
- Mercator map projection, 294, 377
- Mercator, G., 294
- Mercator, N., 133
- meromorphic function, 417
- Merton College, Oxford, xxx, 209
- method of images, 604–611
- metric of a surface, 323
- Midpoint Riemann Sum, 432
- Milne–Thomson Circle Theorem, 608
- Minding, F., 321
- minimal surfaces, 579
- Minimum-Modulus Theorem, 405–406
- Möbius transformations, 137
 as automorphisms of the disc, 199–204
 as matrices, 177–178
 basic results on, 168–177
 classification of, 172
 decomposed into 2 or 4 reflections, 194–199
 decomposed into simpler mappings, 139
 in Einstein's Relativity Theory, 138–139
 in non-Euclidean geometry, 315, 325–329,
 349–353, 356, 360, 363–367, 372
 normalized, 170
 of upper half-plane, 211, 349
 visualized and classified, 184–194
- modular functions, xx
- modular group, xx
- modular surface, 63, 425
 and Cassinian curves, 70, 125
 near essential singularity, 417
 of $\sin(z)$, 95
 of polynomial, 69, 70
 of powers, 63
 poles as “volcanoes”, 74, 417
 related to maxima and minima, 126, 406,
 407, 424
- moments, multipole, 560, 607
- momentum, 541
 angular, 274, 279
- Morera's Theorem, 471
- Morse Theory, 536
- motions
 direct, 39
 Euclidean, 38–41, 373
 hyperbolic, 348–353, 369–373
 isometry, 35
 opposite, 39
 rigid, 35
 spherical, 317–321, 373
- multifunctions, 100–110
- multiplicity
 algebraic, 394–396, 398
 topological, 397–403
- multiplier of Möbius transformation, 186
 calculation of, 191–192
 eigenvalue interpretation of, 192–194
 geometric meaning of, 189–190
- multiply connected set, 103
- multipole, 560
 at infinity, 560
 interpretation of Laurent series, 561
 superposition of multipoles, 561
- multipole expansion, 561
- multipole moments, 560, 607
- mythology, 440
- natural boundary, 299
- Needham
 Claudia, xxxi
 Faith, xiii
 Guy, xxxi
 Hope, xiii
 Mary, xxxi
 Rodney, xxxi
- Needham, T., 276, 278, 280, 631
- Neumann's formula, 637
- Newton
 ultimate equaility, xv–xix
 embraced geometrical methods in 1680s,
 xvi
 Principia, xviii

- priority battle with Leibniz, [xv](#)
- shunned his 1665 calculus in the *Principia*, [xv](#)
- synthetic method of fluxions, [xvi](#)
- Nikulin, V. V. and Shafarevich, I. R., [33](#)
- non-commutative multiplication, [49](#), [330](#)
- non-singular Möbius transformation, [139](#)
- normalized Möbius transformation, [170](#)
- Nyquist Stability Criterion, [422](#)
- octupole, [560](#)
- orbits in force fields, [274–280](#)
- order of branch point, [102](#)
- order of critical point, [232](#)
- orthogonal trajectory, [125](#)
- osculating plane, [298](#)
- Osgood, W. F., [643](#)
- Pappus, [xv](#)
- parabolic Möbius transformation, [172](#), [190](#)
- Parallel Axiom, [304](#)
- parallel transport, [383](#)
- parametric evaluation, [466–467](#)
- partial fractions, [25](#), [455](#)
- Pascal's triangle, [129](#)
- Peaucellier, A., [205](#)
- Penrose, R.
 - 2-spinors, [180](#)
 - and complex numbers in physics, [xxvii](#)
 - and Möbius transformations in relativity theory, [137](#)
 - author's debt to, [xiv](#), [xxx](#)
 - labelling of lights rays with complex numbers, [206](#)
 - on Differential Geometry, [373](#)
 - on quaternions, [333](#)
- Perelman, G., [xx](#)
- periodic function
 - $\cos z$, [98](#)
 - $\sin z$ and $\cos z$, [94](#)
 - exponential function as, [89](#)
 - Fourier series representation, [86](#)
 - saw tooth function, [134](#)
 - square wave function, [134](#)
- Perkins, F. W., [637](#), [643](#)
- Perseus, [70](#)
- perspective drawing, [209](#)
- perspectivity, [36](#)
- phase portrait, [514](#)
- photons, [206](#)
- Pick's curvature formula, [298](#)
- Pick's Theorem, [411](#)
- Pick, G., [411](#)
 - influence on Einstein, [411](#)
- planetary motion, [274–280](#)
- Poincaré, H.
 - and motions of hyperbolic 3-space, [315](#)
 - hyperbolic motions as Möbius transformations, [315](#)
 - motions of hyperbolic 3-space, [372](#)
 - rediscovery of Beltrami hyperbolic models, [xii](#), [342](#), [359](#)
- Poincaré–Hopf Theorem, [526](#), [529–530](#)
- point at infinity, [157](#)
- Poisson's formula
 - for disc, [631](#)
 - for upper half-plane, [637](#)
- polar line of a rotation, [320](#)
- pole, [415](#), [416](#)
 - order of, [417](#)
- pole of a rotation, [320](#)
- Pólya vector field
 - area as flux, [550–551](#)
 - Braden's Theorem, [575](#)
 - Cauchy's formula, [555–556](#)
 - Cauchy's Theorem, [549–550](#)
 - complex integration, [547–549](#), [574](#)
 - definition of, [xxi](#), [547](#)
 - index of, [548](#)
 - integral of z^m , [556–560](#)
 - named by and advocated for
 - by Bart Braden, [xxi](#)
 - non-analytic example, [564](#)
 - of dipole, [557](#)
 - physical meaning of contour integral, [549](#)
 - relation to complex potential, [571](#)
 - winding number as flux, [551–553](#)
- polynomials
 - approximating power series, [78–79](#)
 - Argument Principle for, [393–394](#)
 - Cassian curves via, [68–70](#)
 - cubic, [5](#), [51](#), [66](#)
 - cyclotomic, [57](#)
 - Descartes' factor theorem, [26](#)
 - Fundamental Theorem of Algebra, [26](#), [404](#)

- potential function, 567–569
- power series
 - convergence of, 71–78, 83–86
 - differentiation of, 257–259
 - for $\exp(z)$, 83
 - for $\log(1+z)$, 112–113
 - Laurent, 493–494, 501–505
 - manipulation of, 81–83
 - polynomial approximation of, 78–79
 - related to Fourier series, 86–88
 - Taylor, 87, 491–493
 - uniqueness, 79–80
 - used to find residues, 497–498
- powers of z
 - fractional, 100–106
 - general, 113–115
 - positive integer, 64–66
 - visual differentiation of, 260–262
- primitive root, 57
- principal branch
 - of $\arg(z)$, 105
 - of $\log(z)$, 111
 - of $\sqrt[n]{z}$, 105
 - of $z^{(a+ib)}$, 114
 - used in Binomial Theorem, 113, 130
- principal curvatures, 312, 335, 545
- Principia*, Newton's, xxv, 67, 276, 434
- projections, group of, 36
- projective geometry, 37, 175, 179
- pseudosphere, 307, 334
 - $\neq$ hyperbolic plane, 339
 - area of, 357
 - building your own, 379
 - conformal map of, 336–339
 - constant curvature of, 313, 335–336, 357
 - edge of, 340
 - interpretation of limit rotations, 352
 - radius of, 313
- pseudospherical surfaces, 333, 336, 342
- Ptolemy, C., 156, 158, 207
- quadratics, 68
- quadrupole, 559, 560, 576
 - superposition of multipoles, 561, 562
- quantum mechanics, xxvii
- quaternions, 49, 329–333, 374, 375
- radius of convergence, 71, 73, 74, 78
- recurrence relations, 55, 56
- reflection
 - glide, 58
 - in a circle, 141
 - in a general curve, 289, 290
 - in a hyperbolic line, 345
 - in a sphere, 151
 - in a spherical line, 318
 - Three Reflections Theorem, 42
- reflection property of conics, 99, 100
- reflections, reduction to
 - Euclidean motions, 41
 - hyperbolic motions, 353
 - Möbius transformations, 198
 - spherical motions, 319
- Relativity Theory, Einstein's, 137–139, 379
- repulsive fixed point, 187
- residue
 - evaluation of real integrals, 495–496, 506, 507
 - evaluation of series, 508
 - evaluation of series, 498–501
 - finding using power series, 497–498
 - formula for a simple pole, 494
 - general formula for, 494
- Residue Theorem, 457, 477, 494
- Riemann sphere, 159
 - behaviour of functions at ∞ , 163–164
 - classification of
 - Möbius transformations, 173
 - conformality, 160, 164, 165, 230
 - effect of inversions on, 162, 206, 325–326, 374
 - flows on, 560, 609–610, 650
 - rotations of, 182–184, 325–333
 - symmetry of $\exp(z)$, 206
 - transferring functions to, 162–163
- Riemann sum, 430
 - accuracy of, 434–436
 - complex, 436–440
 - superiority over trapezoidal rule, 436, 478
 - using midpoints, 432
- Riemann surfaces
 - and critical points, 233
 - and doughnuts, 230
 - are two dimensional, 62
 - Further Reading, 530

- omitted from this book, xxvii
- representation of multifunctions, 466
- Riemann's Mapping Theorem, 204, 613–630
- Riemann, B.
 - and birth of complex analysis, 3
 - and Macdonald's Theorem, 424
 - discovered n -dimensional Riemannian geometry, 308
 - doctoral thesis, 204
 - his physical approach to
 - Riemann surfaces, xxvii
 - introduction of Riemann sphere, 158
- rigid motion, 35
- rigidity of analytic functions, 249–252
- robotic vision, 332
- Rodrigues, O., 49, 320
- roots
 - algebraic multiplicity of, 394–396, 398
 - Descartes' Factor Theorem, 26
 - Fundamental Theorem of Algebra, 26, 404
 - of unity, 29
 - primitive, 57
 - Rouché's Theorem, 403
 - simple, 395
 - topological multiplicity of, 397–403
- rotations
 - as complex functions, 19
 - as two reflections, 319, 350
 - composing, 20
 - of space, 317–321, 325–333
- Rouché's Theorem, 403
- Saccheri, G., 304, 305
- saddle points, 425, 518
- Schwarz function, 289, 299, 300
- Schwarz's formula, 637
- Schwarz's Symmetry Principle, 291, 605, 607
- Schwarz, H. A., 289, 631
- Schwarzian derivative, 297, 481
 - and curvature, 298
- Schwarzian reflection
 - becomes reflection near the curve, 300
 - definition of, 289–292
 - in an ellipse, 299
 - used to generalize method of images, 609
 - used to generalize *symmetric circles*, 301
- Shaw, W. T., xix, 139, 180
- Siegel, C. L., 618
- similarities, 44–48
 - direct or opposite, 46
- simple loop, 386
- simple pole, 417
- simple root, 395
- simply connected set, 103
- sine, 94–100
- singular point, 518
 - types of sector surrounding, 532
- Singular Value Decomposition, 236
 - 2-dimensional, 236, 237
 - 3-dimensional, 243
 - eigenvalues, 421
- singular values (of linear transformation), 237
- singularities
 - classification of, 503
 - essential, 417
 - of pole type, 415, 416
 - order of, 417
- sinks, 516–518
 - index of, 519
- soap films, 579
- Soddy, F., 209
- Sommerville, D. M. Y., 237
- source-vortex
 - explanation of Cauchy's Formula, 556
 - role in Laurent series, 561
- source-vortex combination
 - role in Laurent series, 562
 - superposition of multipoles, 561, 562
- sourceless vector field
 - and harmonic duals, 581
 - definition of, 539
 - has harmonic potential
 - if conservative, 569
 - has harmonic stream function if irrotational, 567
 - of complex curvature of analytic mapping, 592
 - properties if irrotational, 545–547
 - stream function of, 563
 - strength proportion to density
 - of k -flux tubes, 564
 - used to prove Green's formula, 650
- sources, 516–518
 - Pólya vector field of $(1/z)$, xxi

- spacetime
 - Lorentz transformations, 138
 - Penrose's stereographic construction, 206
- spherical geometry, 313, 315
 - absolute unit of length, 306
 - angular excess, 316–317
 - angular excess proportional to area, 306
 - conformal map, 321–325
 - curvature of, 307
 - definition of, 305
 - motions of, 317–321
 - rotations as Möbius transformations, 325–329
 - rotations as quaternions, 329–333
 - triangles have positive angular excess, 306
- spiric section of Perseus, 70
- stagnation point, 571
- Star Trek*
 - The City on the Edge of Forever*, xiii
 - Guardian of Forever*, xiii
- Steiner, J., 208
- stereographic projection
 - as inversion in a sphere, 161
 - conformality of, 160, 164, 165, 230
 - formulae for, 165–168
 - in hyperbolic geometry, 367–373
 - in spherical geometry, 324–329
 - of antipodal points, 167
 - Penrose's spacetime construction, 206
 - preservation of circles, 161
- Stewart, I. and Tall, D., xix, 4
- Stiefel Vector Field, 534
- Stiefel, Eduard, 534
- Stillwell, J., xxvii
- Stokes's Theorem, 547
- stream function, 563–565
- streamlines, 514
 - k-flux tube, 563
- strength of a source, 516
- subgroup, 41
- superposition of multipoles, 561, 562
- surface
 - geometry on, 307–309
 - of constant curvature, 313–315
 - vector field on, 525–531
- SVD, 237, 243, 421
- symmetric points, 141
 - preservation under inversion, 149–150
- Symmetry Principle, 168
 - disc automorphisms via, 200–201
 - Schwarz's, 291
 - used to decompose Möbius transformations, 195
- Tartaglia, N., 5
- temperature as potential function, 567, 624
- Three Reflections Theorem, 42
- Thurston, W. P., 336, 361, 372, 379
- Topological Argument Principle, 399–403
- topological multiplicity, 397–403
- topology
 - Euler characteristic, 527
 - Euler-Lhuillier Formula, 534
 - genus of surface, 526
 - Hopf's Degree Theorem, 389
 - Poincaré–Hopf Theorem, 526
 - universal cover, 340
- torus, 526
 - and lemniscate, 70
 - and Riemann surfaces, 230
 - spiric sections of Perseus, 70
 - vector field on, 527
- tractrix, 333
- translations
 - as complex functions, 19
 - composing with rotations, 20
- transplanted function, 583
- transpose (of matrix)
 - simple geometrical interpretation of, 237
- transpose, conjugate, 183
- Trapezoidal Rule, 432
- triangle inequality
 - Braden's Theorem, 575
 - generalized, 8
 - integral version of, 440–441
- trigonometry, 16–17
- triple pole, 417
- twist, 218, 219, 224
 - as verb, 218, 224
- Ultimate Equality
 - $\asymp$ -notation, xvi
 - adoption of $\asymp$ -notation
 - by Chandrasekhar, xvi
 - basic properties of, xvi

- definition, [xvi](#)
- Newton's use of in the
 - Principia*, [xv](#), [xviii](#)
- used to calculate $(\tan \theta)'$, [xvii](#), [xxv](#)
- used to study complex differentiation, [245](#)
- velocity of spiral motion, [23](#)
- ultra-parallel h-lines, [350](#)
- uniform convergence, [79](#)
- uniqueness
 - of analytic continuation, [284–286](#)
 - of Möbius transformations, [170](#)
 - of power series, [79–80](#)
- unitary matrix, [183](#)
- universal cover, [340](#)
- upper half-plane
 - interpretation of cross-ratio, [176](#)
 - Möbius transformations of, [211](#), [349](#)
 - model of hyperbolic geometry, [339–342](#)
 - Poisson's formula for, [637](#)
- valence, [396](#)
- vector
 - sourceless, [547](#)
- vector field, [511](#)
 - conservative, [541](#), [567–569](#)
 - fiducial, [528](#)
 - irrotational, [542](#), [545–547](#), [567](#), [581](#), [600](#), [616](#)
 - on a curved surface, [525–531](#)
 - sourceless, [539](#), [545–547](#), [563](#), [564](#), [567–569](#), [581](#), [592](#), [650](#)
 - Stiefel, [534](#)
- vectorial operations, [30–33](#), [330](#)
- velocity, [12](#)
- Viète's cubic solution, [66](#)
- vortex, [555](#)
 - index of, [519](#)
- Wallis, J., [2](#), [4](#), [481](#)
- Wessel, C., [2](#), [3](#)
- Whyburn, G. T., [429](#)
- winding number, [385](#)
 - and Cauchy's Formula, [488–489](#)
 - and Residue Theorem, [475–477](#), [494](#)
 - as flux, [551–553](#)
 - as integral of $(1/z)$, [446](#)
 - crossing rule for, [388](#)
- work
 - l-work tube, [567](#)
 - circulation interpretation of, [541](#)
 - local density of, [543](#)
 - Stokes's Theorem, [547](#)
 - used to define potential function, [567](#)
 - vanishing of, [xxi](#), [541](#), [542](#), [545](#), [549](#), [557](#)
- Y&Y, Inc., [xiii](#), [xxiv](#)
- Zeitz, P., [xxix](#), [129](#), [154](#)